

Beetle Atlas₂

The *Tribolium* gene expression atlas



Beetle Atlas 2

User's Manual

<https://www.beetleatlas.org>

December 2025

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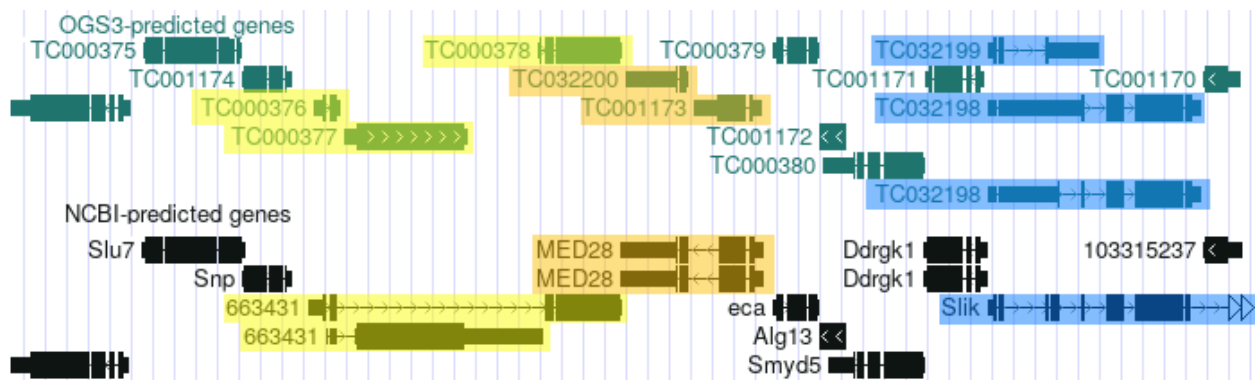
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Introduction

BEETLE ATLAS was conceived as a definitive, tissue-specific transcriptomic resource for the red flour beetle, *Tribolium castaneum*. It is based on RNA-Seq data and, in its simplest form, provides answers to the question “Where is my gene of interest expressed?”. For different tissues of adults and larvae it provides the transcript abundance (FPKM) and enrichment (compared to whole beetle), and for all conditions there is information on gene splice variants (i.e. isoforms). In addition, one can examine the overall expression at different stages of embryonic development. The first part of this document, ‘General Procedures’, describes the web application with an emphasis on the types of search available.

In 2025 BeetleAtlas was updated to adapt it to the new NCBI Reference assembly of the *Tribolium castaneum* genome — icTriCast1.1. This involved reprocessing the original RNAseq data against the new genome sequence and the associated ‘gene models’ (redicted gene assignments — see further, below) which are referenced by numerical NCBI identifiers. This posed a problem in relation to the preceeding version of BeetleAtlas which, in common with the majority of work in the field, used the OGS3 (Official Gene Set 3) gene models based on the previous Tcas5.2 assembly, and had a different set of identifiers (TC012345 etc.). We initially assumed that although the new NCBI gene set might be of higher quality than the OGS3 gene set, there would be substantial overall agreement between the two. Thus, all that would be needed would be to provide correlations between the gene models in the two sets and incorporate them into the underlying database, enabling searches with OGS3 gene identifiers as well as NCBI identifiers in most cases. Unfortunately such correlation only proved possible in about 60% of cases, and further study showed this reflected extensive differences between the two gene sets. Furthermore, where there was conflict between NCBI and OGS3 gene models our own analysis indicated that, overall, neither was more likely to be correct.



Example of region of the *T.castaneum* genome with several differences between OGS3 and NCBI gene models.

In consequence *BeetleAtlas2* has become a major update, embodying data processed using both gene model sets and a facility for checking results and resolving disparities. There are now:

- Separate portals for searching by OGS3 or NCBI identifiers
- A custom UCSC Genome Browser view of the gene models and RNAseq reads

The second part of this document explains how to use the new UCSC Genome Browser view. Please read it.

The results from BeetleAtlas searches are only as good as the underlying gene models.

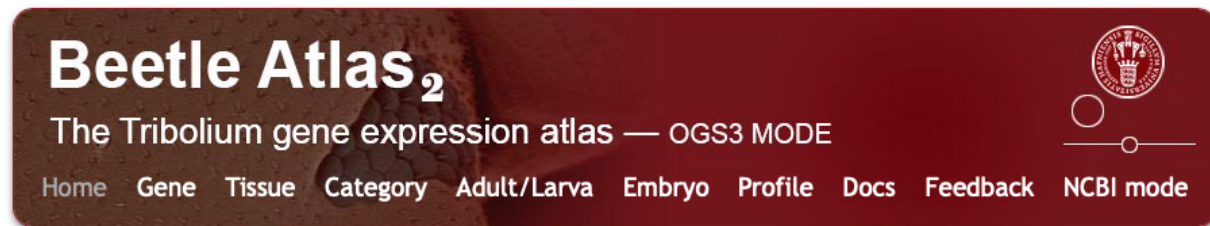
General Procedures

Mode Selection

On launching BeetleAtlas 2 you are required to select either OGS3 or NCBI modes. If your gene of interest has an alpha-numeric identifier of the type TC012345, choose OGS3, if it has a numeric identifier choose NCBI. Both modes have a similar main menu and allow one to switch between them.

OGS3 Mode

NCBI Mode



Search Types

The menu presents several different search options:

- GENE: For a particular gene, find details of its expression in different tissues or in embryos.
- TISSUE: For a given tissue, find the genes of a particular category with the greatest expression.
- CATEGORY: For a given tissue, find highly expressed genes corresponding to certain assigned keywords.
- ADULT/LARVA: For a given tissue, find the genes that show different expression between larva and adult.
- EMBRYO: Find genes expressed at different stages of embryo development.
- PROFILE: Find genes with a pattern of expression similar to a gene of interest.

(i) Gene Search

Searching for tissue-specific expression data for a single gene is quite straightforward: select the type of input, enter the search term, click on the correct autocomplete* text and hit the search button. One can search using *Drosophila* identifiers as well as those for *Tribolium*.

For a particular *Tribolium* gene, find the pattern of expression in different tissues.

• Choose search type, start entering text, then select from the autosuggest menu •

- ☒ Gene ID (e.g. TC006446)
- ☐ Gene Symbol (e.g. Hcf)
- ☐ Gene Product (enter partial term like 'eye', 'inositol')
- ☐ NCBI ID (switches modes)

or use a *Drosophila* ID to search for related *Tribolium* genes

- ☐ FlyBase ID (e.g. FBgn0016075)
- ☐ FlyBase CG number (e.g. CG16858)
- ☐ FlyBase Symbol (e.g. vkg)

Gene:

*Using the autocomplete will save yourself the frustration of making a query, only to find that there are no results. If the gene is not in the autocomplete menu you can be sure that it is not in the database! If you think it should be, check for typos or that you have the appropriate search type selected. N.B. Autosuggest for a Symbol starts after two characters, and for a Gene ID after six.

(ii) Tissue Search

Find which genes have the greatest expression in a particular tissue.

First select a 'stage' and next the tissue of interest. Then choose 'enrichment' or 'abundance' for greatest expression:

Adult Hindgut Enrichment

Display: 30 Search

ANATOMICAL KEY show

Tissue search allows one to search for the genes with the greatest expression in a particular tissue. First one chooses the tissue — there is a pop-up anatomical guide to assist you, if required. Then one selects whether one wishes to see the genes with the greatest absolute expression (abundance) or those which are most highly expressed relative to other tissues (enhancement). Finally one selects how many results one wishest to see (20–100).

(iii) Category Search

Find genes of a particular category most expressed in individual tissues.

1. Select a 'stage' and next the tissue of interest. Then choose 'enrichment' or 'abundance' for greatest expression:

Larval Tubule Enrichment

2. Select a category (keyword), how many genes you wish displayed, and press 'Search'.

Category: ion channel Display: 40 Search

ANATOMICAL KEY show

Category search offers a pull-down list from which one can select any of 24 functional categories for gene products, and allows one to see which are expressed in a particular tissue.

(iv) Adult/Larvae Search

Find which genes show the greatest difference in expression between larva and adult for a given tissue.

1. Select whether expression should be greater in the adult or larval stage:

☐ Adult > Larval
☒ Larval > Adult

2. Choose the tissue, how many genes you wish displayed, and press 'Search'.

Tissue: Fat Body Display: 50 Search

ANATOMICAL KEY show

This type of search simply allows one to find genes in a particular tissue that are much more highly expressed either in adult or in larvae.

(v) Embryo Search

Find genes expressed in the embryo.

Select a peak of embryo expression.

☐ 0–1 h

☐ 1–24 h

☒ 24–36 h

☐ 36–72 h

Do you wish to see genes expressed only in embryo?

☒ Yes ☐ No

Display:

The embryo search allows one to identify genes, the expression of which is predominantly at a particular embryonic stage, which one chooses from the choices offered. There is the option to restrict the choice of genes to those that are only expressed in the embryo.

(vi) Profile Search

For a particular *Tribolium* gene, find others with a similar profile of expression across tissues.

Choose search type before entering ID or symbol

☐ Gene ID (e.g. TC000470)

☒ Gene Symbol (e.g. Sobp)

Include embryo data in profile?

☐ Embryo/Larva/Adult ☒ Larva/Adult only

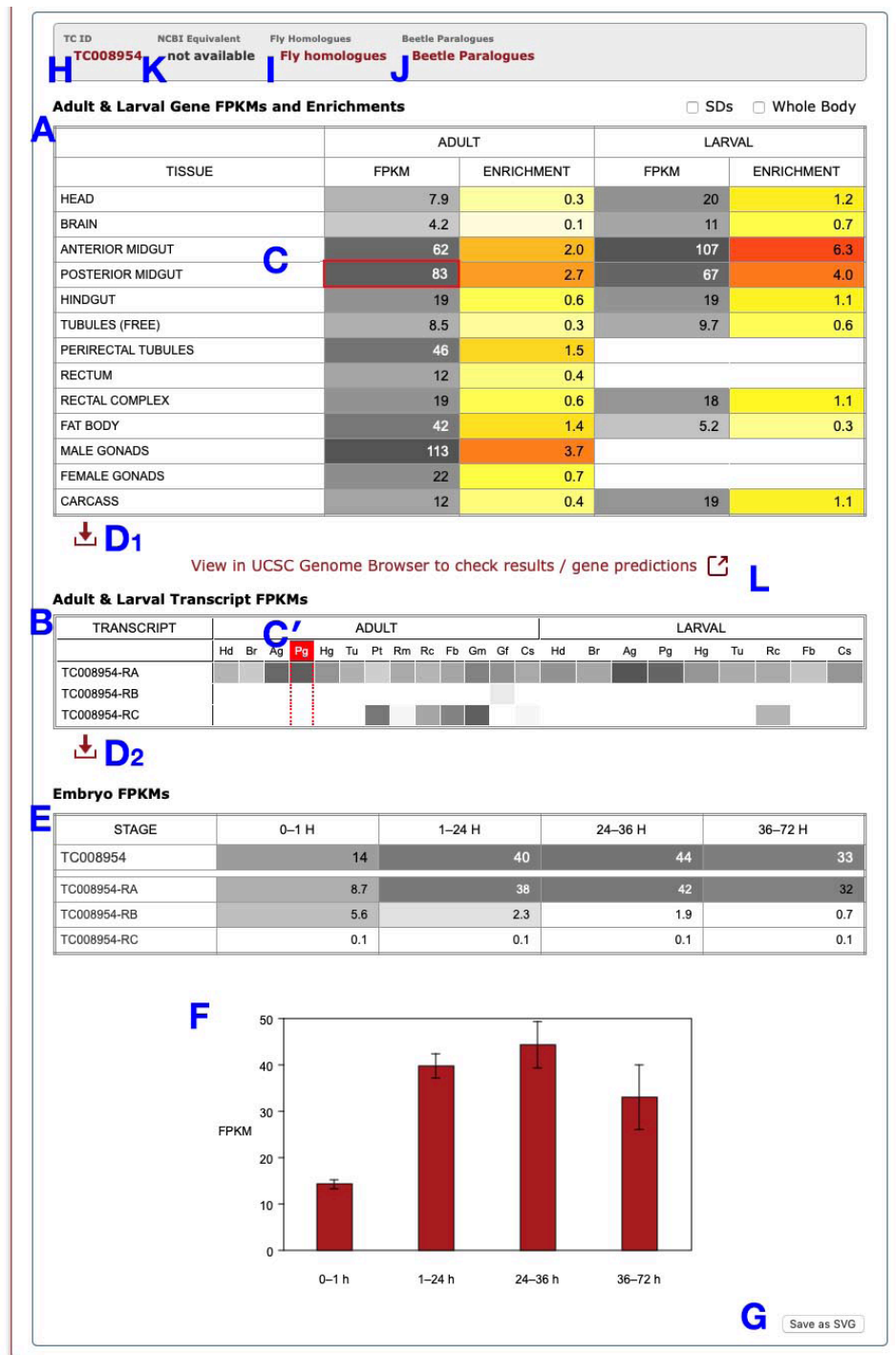
Correlation statistic and stringency (Defaults advised initially)

☒ Pearson ☐ Spearman $r > 0.70$

Gene: Display:

The purpose of the profile search is to help identify the function of genes for which this is unknown. The search employs the pattern of expression of a query gene across different tissues and retrieves genes with a similar pattern of expression. The results one obtains may differ depending on the statistical model used for the comparison, and one can select either Pearson or Spearman statistics. Likewise, there is an option to exclude comparison with expression in the embryo. It is suggested that one experiments with these options to obtain a suitable number of results to a particular query. If one obtains a large number of genes with a similar expression profile, there is the option to increase the stringency of the comparison by selecting a higher r value.

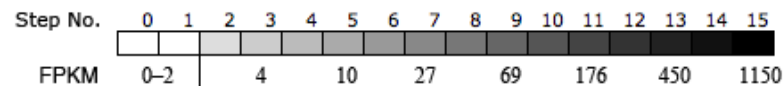
Presentation of Results



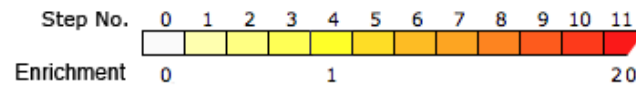
Adult and Larval Expression

The overall data for gene transcripts in different tissues are tabulated first (A), and below this those for individual transcripts in a more compact form (B). It is possible to locate the expression of transcripts corresponding to that of the gene in particular tissue by clicking on the gene cell (C), causing the corresponding transcript cells (C') to be highlighted. The data can be downloaded in spreadsheet format (D₁ and D₂).

For abundance values the white-to-black spectrum illustrated below is used. For FPKMs this has a 15-step log1.6 scale, but all values below 2 FPKM (considered background) are coloured white, and no discrimination is employed for values above 1150 FPKM (pure black).



For enrichment values a divergent white/yellow/red spectrum is used:



This employs an asymmetric log scale running from white to yellow (0-1) to represent genes with ‘decreased enrichment’, and then from yellow through orange and a wider range of reds to encompass values of ‘increased enrichment’.

Embryonic Expression

Embryonic expression is presented in the form of both a single table for both gene and individual transcripts (E) and a bar chart of the overall gene expression (F). The latter can be downloaded in scaleable vector graphics (SVG) format (G)).

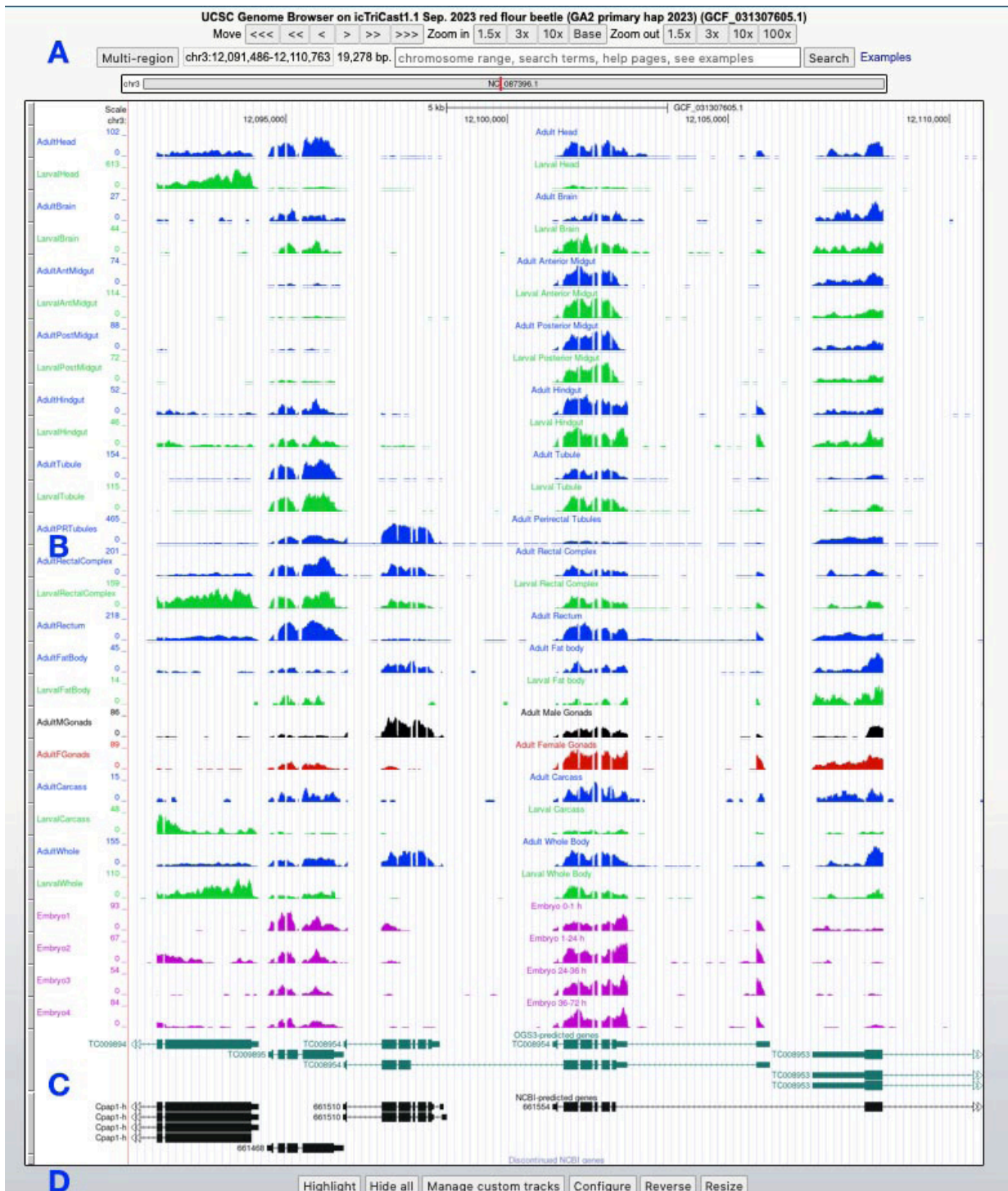
Links

For each retrieved gene there are links out to the corresponding entry at iBeetleBase or NCBI, as appropriate (H). Links are also provided to Drosophila homologues (I) and Tribolium paralogues (J) — where they exist. Of particular importance is the link (K) to any equivalent gene for the alternative mode — here the NCBI mode — in this case absent. This absence makes it especially important to examine the gene model underlying these results using the link to the custom UCSC Genome Browser view (L). This is described in the following section.

Using the UCSC Genome Browser view

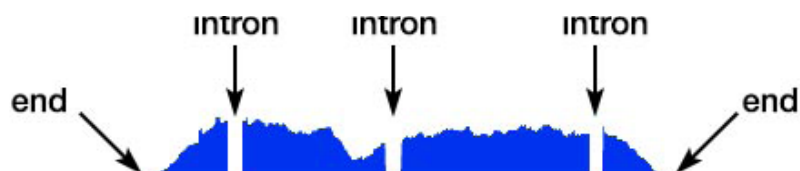
General

On invoking the link (L above) to the custom UCSC Genome Browser, one is eventually presented with a view of the type shown below. General information and controls are presented at the top (A), with the Zoom controls being particularly useful. The main display area shows the RNAseq coverage for different tissues (B) in relation to the OSG3 and NCBI gene models (C), for which more information is available on 'mouse-over'. Further controls (D) are placed outside the display area, including the ability to reverse genome direction and to display additional tracks such as the nucleotide sequence. The Browser is very powerful, and it is worth taking the time to learn about it.



Relating custom tracks to the gene models

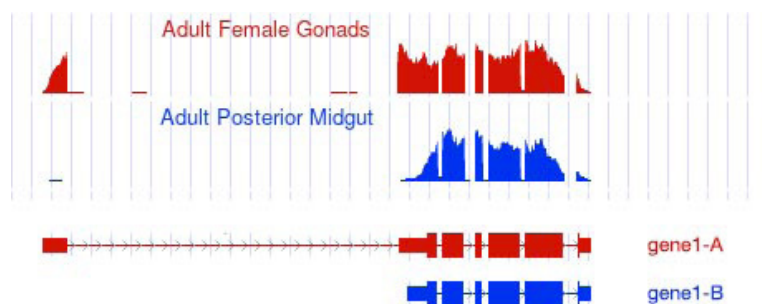
A forthcoming publication provides a detailed description of how to use the custom view in the UCSC Genome Browser to check the validity of the results presented in BeetleAtlas2 and to confirm, improve upon or reject individual gene models. Here we will merely summarize our approach and present an example. In general the problem is determining the position of the 3' and 5' ends of a gene model — frequently to distinguish between a single gene and two separate genes. The appearance of the RNAseq reads is key here: abrupt interruptions ('cliff edges') indicate introns (and hence continuation) whereas a gradual decline is found at the ends of the transcript. Comparison of the coverage in different tissues can often also be of assistance: if two parts of a putative gene model are only expressed in different tissues, an alternative model with separate genes would be favoured.

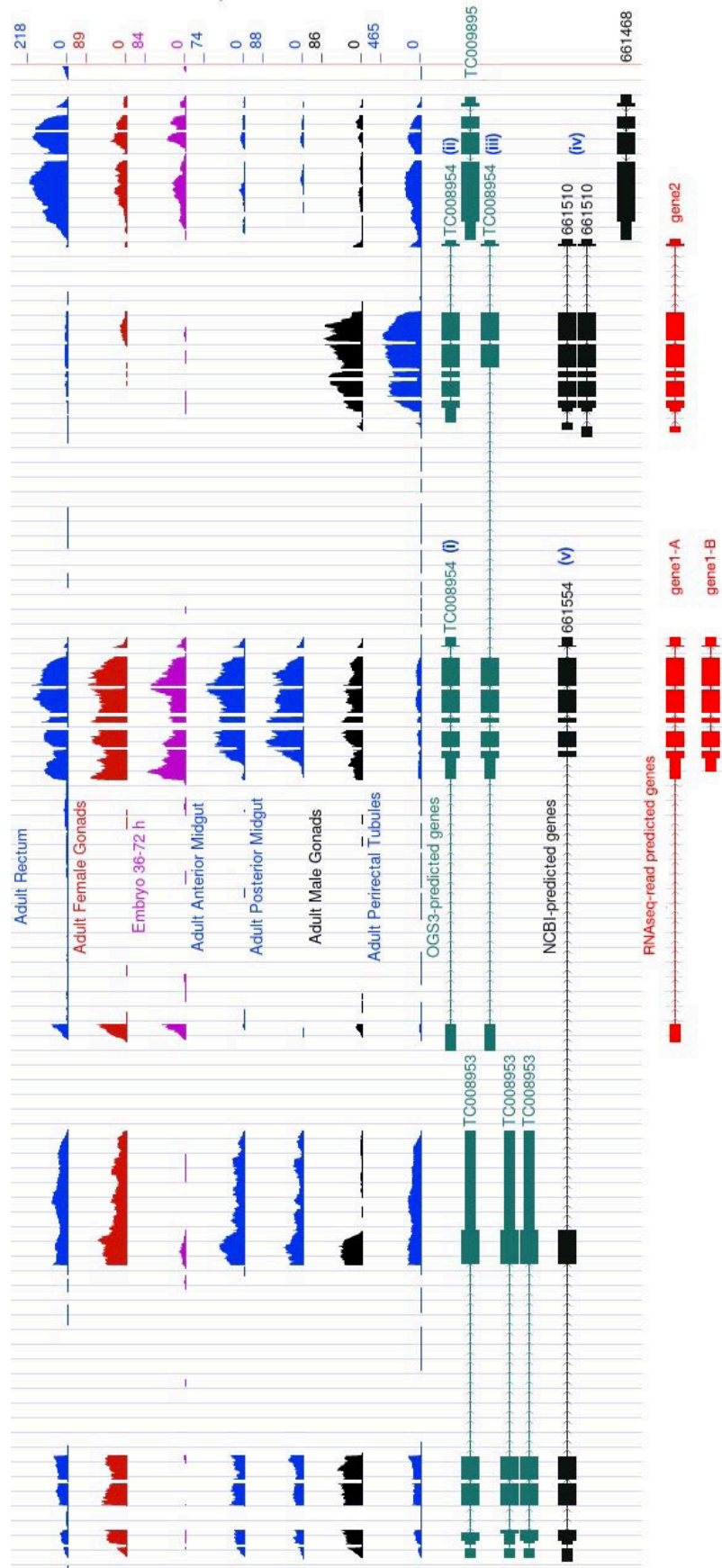


As an example we shall consider gene model TC008954, used above to illustrate the way in which results are presented. The reader will recall that no 'NCBI equivalent' was flagged in those results, something that surprised the author who had used the gene previously for version 1 of the manual, where its product was described as beta-1,3-galactosyltransferase-1. One might expect such a gene to have been characterised in the NCBI gene set.

The invoked UCSC Genome Browser view shown above is clearly quite complex, and it was adjusted to bring representative tracks of most interest closer to the gene models by dragging them by their handles. A view of the adjusted tracks (zoomed and scrolled appropriately and with the genome direction reversed) is shown opposite. To this has been added the author's alternative proposals for gene models in this area, which were arrived at as follows:

- The OSG3 model shows three transcripts for TC008954, two (i and ii) quite distinct, but presumably classified as part of the same gene because transcript (iii) is predicted to span both. The RNAseq traces for several tissues (anterior and posterior midgut, specifically) are consistent with (i), and male gonads and perirectal tubules with (ii), but provide no support for (iii), which is not predicted in the NCBI model set. We therefore regard (i) and (ii) as separate genes which are designated gene1 and gene2 in the diagram.
- The NCBI model set predicts a gene model, 661510 (iv), which corresponds to (ii)/gene2 and is actually annotated as beta-1,3-galactosyltransferase-1. The RNA seq data appear to correspond to one of the two isoforms presented.
- The NCBI model set does not predict a gene model corresponding to that we have designated gene1, but instead predicts the longer 661514 (v), encompassing it and the OSG3 model TC008953. The RNAseq data are consistent with TC008953, but provide no support for 661514.
- Close examination of the RNAseq reads in the region of gene1 indicate two isoforms, A and B: the latter lacking the first exon, and representing transcription from an alternative initiation site.





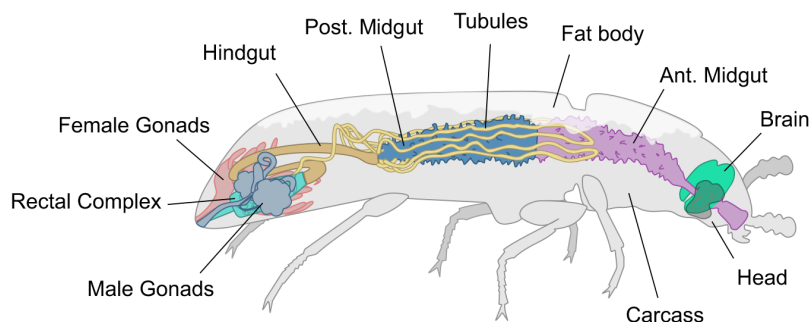
Improving the annotation of the genome of *T. castaneum*

The foregoing example highlights both the deficiencies of the current gene model sets for *T. castaneum* and the utility of the custom UCSC Genome Browser view of BeetleAtlas2 in correcting them. At the moment computerized gene prediction is insufficient — human intervention is required. Although most users will be concerned with a few specific genes of relevance to their own research, we would ask them to contribute any discrepancies they discover to help produce a revised genome annotation. Currently we suggest using the Feedback Page to do this, although we hope to have a standardized means of doing so in the future.

Appendix I: Technical

Insects & Tissues

Tribolium castaneum (San Bernardino strain) stocks were maintained on organic whole-wheat flour supplemented with 5% (w/w) yeast powder at 30 °C at a constant 50% relative humidity and 12:12 light–dark cycles. The tissues are in the ‘Anatomical key’, available on each results page and reproduced below.



RNA-Seq

RNA was isolated in triplicate using the Qiagen miRNeasy kit. The quality of RNA samples was analysed using an Agilent Bioanalyzer.

Data Analysis

Total RNA was analysed using the Tuxedo pipeline (Trapnell *et al.* (2012) Nature Protocols **7**, 562–578). Two separate ‘runs’ were performed. The first — for the ‘OGS3’ mode of the web application — used as reference the Tcas5.2 genome assembly (https://www.ncbi.nlm.nih.gov/assembly/GCF_000002335.3/) and the OGS3 annotations (<http://ibeetle-base.uni-goettingen.de/download/OGS3.gff.gz>), as for the original BeetleAtlas. The second — for the ‘NCBI’ mode of the web application — used as reference the icTriCast1.1 genome assembly and the NCBI annotations (https://genome.ucsc.edu/h/GCF_031307605.1). The processed data, as FPKM (Fragments Per Kilobase of transcript per Million mapped reads) were deposited in two separate databases — one for OGS3 mode, and one for NCBI mode — and these are the basis for the tables returned by BeetleAtlas 2 in response to queries in the two different modes. Data for small RNAs were not included in the database as the 100bp RNA-seq reads do not detect them. Data for mitochondrial transcripts were also excluded.

Customization of the UCSC Genome Browser

The tissue-specific transcript coverage files for display in the UCSC Genome Browser were generated from ‘BAM’ files which are an intermediary in processing the raw RNA-seq reads through the Tuxedo pipeline. The reference gene model sets were generated as ‘BigGenePred’ files from the annotation files indicated above, but in the case of the OGS3 model set the co-ordinates were first transformed from those of the Tcas5.2 assembly to those of the icTriCast1.1 genome assembly using the program ‘Liftoff’. An additional track was also generated for NCBI gene models from the Tcas5.2 assembly that had been discontinued in the annotation of the icTriCast1.1 assembly.

Appendix II: Citing, Linking and Contact

Authorship & Acknowledgements

The dataset underlying this web application was generated by Kenneth V. Halberg and Muhammad Naseem, University of Copenhagen, Denmark. The web application and the underlying database is by David P. Leader, University of Glasgow, UK. The ‘liftover’ of co-ordinates between genome assemblies was performed by Janina Rinke of the University of Münster, Germany.

Citation in Publications

Any published work relying on the data here should cite the URL of the website (www.beetleatlas.org). The original BeetleAtlas was described in J Mol Biol. 2024;436(17):168520. A manuscript describing BeetleAtlas 2 has been submitted for publication (December 2025).

Linking to BeetleAtlas

You are welcome to link to BeetleAtlas from your own website.

If you wish to link directly to the results of a search for a specific gene you need to provide the two parameters, GENE_ID and ID_TYPE, in a query of the type:

https://motif.mvls.gla.ac.uk/VERSION/index.html?search=gene&gene=GENE_ID&idtype=ID_TYPE

Examples for OGS3 MODE:

<https://motif.mvls.gla.ac.uk/BeetleAtlas/index.html?search=gene&gene=TC034001&idtype=geneID>

<https://motif.mvls.gla.ac.uk/BeetleAtlas/index.html?search=gen&gene=Hcf&idtype=geneSymbol>

Examples for NCBI MODE:

<https://motif.mvls.gla.ac.uk/BeetleAtlas2/index.html?search=gene&gene=657767&idtype=ncbiID>

<https://motif.mvls.gla.ac.uk/BeetleAtlas2/index.html?search=gene&gene=Hcf&idtype=ncbiSymbol>

Contact

Bug reports, feedback and feature requests are welcomed. There is a feedback form on the website, or you may email us directly:

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